

Appendix A

Calculus of Variations

We can think of a function $y(x)$ as being an operator that, for any input value x , returns an output value y . In the same way, we can define a functional $F[y]$ to be an operator that takes a function $y(x)$ and returns a scalar value F . An example of a functional is the length of a curve drawn in a two-dimensional plane in which the path of the curve is defined in terms of a function. In the context of machine learning, a widely used functional is the entropy $H[x]$ for a continuous variable x because, for any choice of probability density function $p(x)$, it returns a scalar value representing the entropy of x under that density. Thus the entropy of $p(x)$ could equally well have been written as $H[p]$.

A common problem in conventional calculus is to find a value of x that maximizes (or minimizes) a function $y(x)$. Similarly, in the calculus of variations we seek a function $y(x)$ that maximizes (or minimizes) a functional $F[y]$. That is, of all possible functions $y(x)$, we wish to find the particular function for which the functional $F[y]$ is a maximum (or minimum). The calculus of variations can be used, for instance, to show that the shortest path between two points is a straight line or that the maximum entropy distribution is a Gaussian.

Using a Taylor expansion, we can characterize the conventional derivative dy/dx by making a small change ϵ to the variable x and expanding $y(x + \epsilon)$ about x :

$$y(x + \epsilon) = y(x) + \frac{dy}{dx}\epsilon + O(\epsilon^2). \quad (\text{A.1})$$

and finally taking the limit $\epsilon \rightarrow 0$. Similarly, for a function of several variables $y(x_1, \dots, x_D)$, the corresponding partial derivatives are defined such that

$$y(x_1 + \epsilon_1, \dots, x_D + \epsilon_D) = y(x_1, \dots, x_D) + \sum_{i=1}^D \frac{\partial y}{\partial x_i} \epsilon_i + O(\epsilon^2). \quad (\text{A.2})$$

The analogous definition of a functional derivative arises when we consider how much a functional $F[y]$ changes when we make a small change $\epsilon\eta(x)$ to the function $y(x)$, where $\eta(x)$ is an arbitrary function of x , as illustrated in Fig. A.1. We denote the functional derivative of $F[y]$ with respect to $y(x)$ by $\delta F / \delta y(x)$, and define it by the following relation:

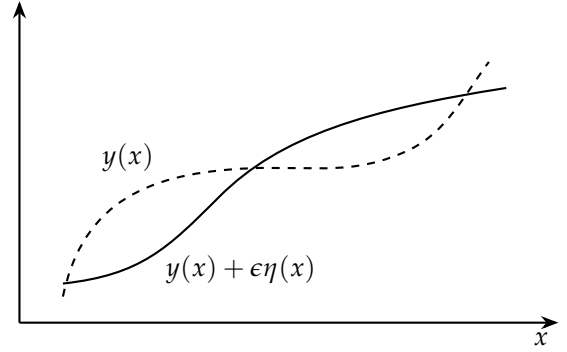


Figure A.1: A functional derivative can be defined by considering how the value of a functional $F[y]$ changes when the function $y(x)$ is changed to $y(x) + \epsilon\eta(x)$, where $\eta(x)$ is an arbitrary function of x .

$$F[y(x) + \epsilon\eta(x)] = F[y(x)] + \epsilon \int \frac{\delta F}{\delta y(x)} \eta(x) dx + O(\epsilon^2). \quad (\text{A.3})$$

This can be seen as a natural extension of A.2 in which $F[y]$ now depends on a continuous set of variables, namely the values of y at all points x .

If our goal is to find a function $y(x)$ that maximizes or minimizes the functional $F[y]$, we require $F[y]$ to be stationary at the optimal function. Here, stationarity means that an arbitrary small perturbation $y(x) \rightarrow y(x) + \epsilon\eta(x)$ produces no first-order change in $F[y]$. From Eq. A.3, the coefficient of ϵ must therefore vanish, giving

$$\int \frac{\delta F}{\delta y(x)} \eta(x) dx = 0. \quad (\text{A.4})$$

Because this condition must hold for every possible perturbation $\eta(x)$, the functional derivative itself must be zero for all x . To see why, suppose that $\delta F / \delta y(x)$ were nonzero near some point $\hat{x}$. We could then choose $\eta(x)$ to be zero everywhere except in a small neighbourhood of $\hat{x}$, and choose its sign so that

$$\frac{\delta F}{\delta y(x)} \eta(x) > 0 \quad (\text{A.5})$$

throughout that neighbourhood. The integral in Eq. A.4 would then be positive rather than zero, which contradicts the stationarity condition. Therefore, $\delta F / \delta y(x)$ cannot be nonzero near any point, and we must have $\delta F / \delta y(x) = 0$ for all values of x .

Consider a functional that is defined by an integral over a function $G(y, y', x)$ that depends on both $y(x)$ and its derivative $y'(x)$, as well as directly on x ,

$$F[y] = \int G(y(x), y'(x), x) dx. \quad (\text{A.6})$$

If we perturb the function according to $y(x) \rightarrow y(x) + \epsilon \eta(x)$, then its derivative is also perturbed according to $y'(x) \rightarrow y'(x) + \epsilon \eta'(x)$. Thus, the first two arguments of G are perturbed, while the independent variable x itself remains unchanged. For small ϵ , we can therefore apply a multivariable Taylor expansion to the integrand,

$$G(y + \epsilon \eta, y' + \epsilon \eta', x) = G(y, y', x) + \epsilon \frac{\partial G}{\partial y} \eta(x) + \epsilon \frac{\partial G}{\partial y'} \eta'(x) + O(\epsilon^2). \quad (\text{A.7})$$

The general solution to this second-order differential equation is

Substituting this expansion of G into the perturbed functional gives

$$\begin{aligned} F[y + \epsilon \eta] &= \int G(y + \epsilon \eta, y' + \epsilon \eta', x) dx \\ &= \int \left[G(y, y', x) + \epsilon \frac{\partial G}{\partial y} \eta(x) + \epsilon \frac{\partial G}{\partial y'} \eta'(x) + O(\epsilon^2) \right] dx \\ &= F[y] + \epsilon \int \left[\frac{\partial G}{\partial y} \eta(x) + \frac{\partial G}{\partial y'} \eta'(x) \right] dx + O(\epsilon^2). \end{aligned} \quad (\text{A.8})$$

There is no term involving $\partial G / \partial x$ because the perturbation changes the function $y(x)$, not the independent variable x .

We now have to cast this in the form Eq. A.3. To do so, we need to express the first-order change in $F[y]$ entirely in terms of $\eta(x)$. However, the second term in the integral contains $\eta'(x)$. We therefore apply integration by parts,

$$\int \frac{\partial G}{\partial y'} \eta'(x) dx = \left[\frac{\partial G}{\partial y'} \eta(x) \right]_{\text{boundary}} - \int \frac{d}{dx} \left(\frac{\partial G}{\partial y'} \right) \eta(x) dx. \quad (\text{A.9})$$

Assume that we are only searching over all admissible functions $y(x)$ that connect fixed endpoint values. The perturbed function $y(x) + \epsilon \eta(x)$ must therefore have the same endpoint values as the original function $y(x)$. This requires $\eta(x) = 0$ at the boundaries, so the boundary term vanishes, leaving

$$\int \frac{\partial G}{\partial y'} \eta'(x) dx = - \int \frac{d}{dx} \left(\frac{\partial G}{\partial y'} \right) \eta(x) dx. \quad (\text{A.10})$$

Substituting this result into the first-order expansion of $F[y + \epsilon \eta]$ gives

$$F[y + \epsilon \eta] = F[y] + \epsilon \int \left[\frac{\partial G}{\partial y} - \frac{d}{dx} \left(\frac{\partial G}{\partial y'} \right) \right] \eta(x) dx + O(\epsilon^2). \quad (\text{A.11})$$

This expression now has the same form as Eq. A.3, so the quantity in square brackets can be identified as the functional derivative $\delta F / \delta y(x)$.

To find a function $y(x)$ at which the functional $F[y]$ is stationary, we require its functional derivative to be 0,

$$\frac{\partial G}{\partial y} - \frac{d}{dx} \left(\frac{\partial G}{\partial y'} \right) = 0, \quad (\text{A.12})$$

which are known as the *Euler–Lagrange equation*. For example, if

$$G = y(x)^2 + (y'(x))^2 \quad (\text{A.13})$$

then the Euler–Lagrange equations take the form

$$y(x) - \frac{d^2 y}{dx^2} = 0. \quad (\text{A.14})$$

$$y(x) = C_1 e^x + C_2 e^{-x}, \quad (\text{A.15})$$

where the constants C_1 and C_2 are determined by the boundary values of $y(x)$.

Often, we consider functionals whose integrands take the form $G(y, x)$ and therefore do not depend on derivatives of $y(x)$. In this case, the Euler–Lagrange equation simplifies directly, and stationarity requires $\partial G / \partial y = 0$ for all values of x .

The extension to a multidimensional variable $\mathbf{x}$ is straightforward by applying the same reasoning componentwise. A small perturbation is now applied to a function $y(\mathbf{x})$, and ordinary derivatives with respect to x are replaced by the corresponding partial derivatives with respect to the components of $\mathbf{x}$. Stationarity is then obtained by requiring the first-order change in the functional to vanish for arbitrary perturbations.